CS 423
Operating System Design: Condition Variables and Semaphores
Mar 10

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Logistics

MP2 out; Due (3/23) 11:59 pm CT

4Cr: No paper for next week

3/12 lecture: The 2 TAs will present 1 topic each

EXPECTATION: That students read OSTEP chapters

AGENDA / LEARNING OUTCOMES

Last class:

- Critical Sections & Locks
- How are locks implemented?
  - Hardware: atomic instructions

Today:

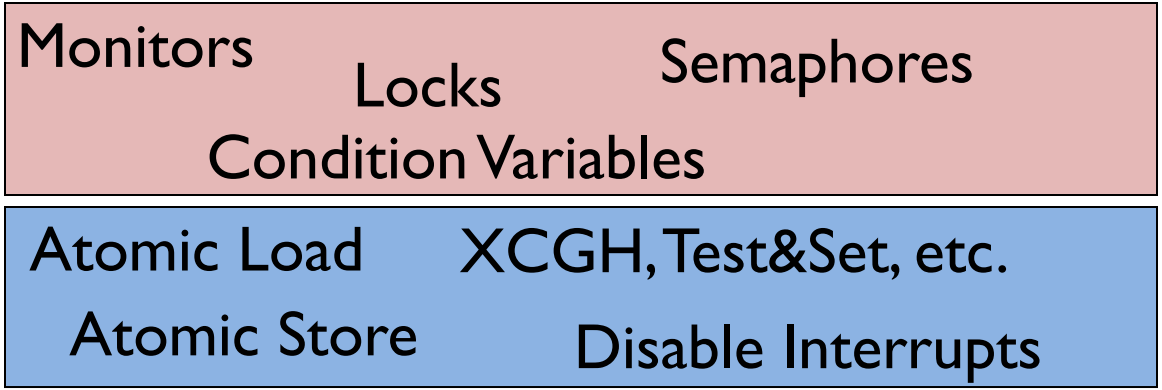
- Condition variables
  - Why do we need them? How are they implemented?
- Solving the Producer-Consumer problem

RECAP

Synchronization

Generic synchronization: for all models from previous slides

Goal: Learn how synchronization primitives are built (lib + OS)
With a little help from our friend…the hardware



LOCKING

Eval Criteria: Correctness, Progress, Bounded waits, Fairness, and Performance.

Implementation (evolution):

- Enable/disable interrupts
- Spin with load/store
- Atomic operations (XCHG)
- Block/sleep instead of spin on lock
- Add a queue: blocked threads get enqueued

Spin-Wait vs Block/Sleep?

Time to context-switch? C
How long before lock is released? T

What is the best action when t < C? spin
When t > C? block/sleep
Problem: Requires knowledge of future

Hybrid approach: Two-phase Locks (not the distributed systems version!)

- Spin for some time (< C)
- Block after that

WHERE IS THE CODE?

User-space (pthread code)

https://codebrowser.dev/glibc/glibc/nptl/

pthread_mutex_[un]lock.c

pthread_cond_{wait|signal}.c

Within the Linux kernel:

/include/linux/mutex.h & /kernel/locking/mutex.c

SPINLOCKs: /include/linux/spinlock.h & /kernel/locking/spinlock.c

But why spin? In very special cases, for very short durations, eg. interrupt handler

Uses semaphores instead of condition variables (we'll discuss those later)

/include/linux/semaphore.h, /kernel/locking/semaphore.c

Concurrency Objectives

Mutual exclusion (A and B don't both run critical section)

- solved with *locks*

Ordering (B runs only after A has done its thing)

- solved with *condition variables* and *semaphores*

END RECAP

ORDERING EXAMPLE: JOIN

```
pthread_t p1, p2;

pthread_create(&p1, NULL, mythread, "A");
pthread_create(&p2, NULL, mythread, "B");
// join waits for a pthread to exit
pthread_join(p1, NULL);
pthread_join(p2, NULL);

printf("main: children threads are done\n”);
return 0;
```

how to implement join()?

Condition Variables

Condition Variable: FIFO Queue of waiting threads

B waits for a signal on CV before it can run

wait(CV, …)

don't spin while waiting

A sends signal to CV when it's time for ***B*** to run

signal(CV, …)

CONDITION VARIABLES

wait(cond_t *cv, mutex_t *lock)

- the lock must be held before calling wait()
- caller is put to sleep & the lock is released (atomically)
- caller is added to a FIFO queue—so fairness is guaranteed.
- returns when awoken, with lock reacquired

signal(cond_t *cv)

```
  if (1+ threads are waiting)
      wake up a thread
 else do nothing (aka no-op, aka NOP)
```

JOIN IMPLEMENTATION: ATTEMPT #1

Parent:

```
void thread_join() {
        mutex_lock(&m);        // P1
        cond_wait(&c, &m);   // P2
        mutex_unlock(&m);   // P3
}
```

Child:

```
void thread_exit() {
        mutex_lock(&m);        // C1
        cond_signal(&c);        // C2
        mutex_unlock(&m);   // C3
}
```

Example schedule:

Parent:	P1	P2				P3
Child:			C1	C2	C3	

Global vars:

```
mutex_t  m;
cond_t   cv;
```

JOIN IMPLEMENTATION: ATTEMPT #1

Parent:

```
void thread_join() {
        mutex_lock(&m);        // P1
        cond_wait(&c, &m);   // P2
        mutex_unlock(&m);   // P3
}
```

Child:

```
void thread_exit() {
        mutex_lock(&m);        // C1
        cond_signal(&c);        // C2
        mutex_unlock(&m);   // C3
}
```

Example broken schedule:

Global vars:

```
mutex_t  m;
cond_t   cv;
```

RULE OF THUMB 1

Need to keep some state in addition to CV's!

CV's are used to signal threads when that state changes

If state is already as needed, thread doesn't wait for a signal!

JOIN IMPLEMENTATION: ATTEMPT #2

Parent:

```
void thread_join() {
        mutex_lock(&m);             // P1
        if (done == 0)              // P2
                cond_wait(&c, &m);  // P3
        mutex_unlock(&m);           // P4
}
```

Child:

```
void thread_exit() {
        done = 1;          // C1
        cond_signal(&c);// C2
}
```

Fixes previous broken schedule

Parent:		P1	P2	P3	P4
Child:	C1	C2			

Global vars:

```
mutex_t  m;
cond_t   cv;
boolean done = 0;
```

JOIN IMPLEMENTATION: ATTEMPT #2

Parent:

```
void thread_join() {
        mutex_lock(&m);             // P1
        if (done == 0)              // P2
                cond_wait(&c, &m);  // P3
        mutex_unlock(&m);           // P4
}
```

Child:

```
void thread_exit() {
        done = 1;          // C1
        cond_signal(&c);// C2
}
```

An example broken schedule:

Global vars:

```
mutex_t  m;
cond_t   cv;
boolean done = 0;
```

JOIN IMPLEMENTATION: CORRECT

Parent:

```
void thread_join() {
        mutex_lock(&m);              // P1
        if (done == 0)               // P2
                cond_wait(&c, &m);   // P3
        mutex_unlock(&m);            // P4
}
```

Child:

```
void thread_exit() {
        mutex_lock(&m);      // C1
        done = 1;            // C2
        cond_signal(&c);     // C3
        mutex_unlock(&m);    // C4
}
```

Parent: P1 P2 P3 P4

Child: C1 C2 C3 C4

Global vars:

```
mutex_t  m;
cond_t   cv;
boolean done = 0;
```

Use mutex to ensure no race between interacting with state and wait/signal

CV RULE OF THUMB 2

1. Modify/check state with mutex held
2. Mutex is required to ensure state doesn't change between checking the state and waiting on CV
3. Always use a while instead of an if when checking cond
   a. To avoid getting fooled by "spurious wakeups"

```
mutex_lock(&m);
while (done == 0) {
    cond_wait(&c, &m);
}
mutex_unlock(&m);
```

```
mutex_lock(&m);
if (done == 0) {
    cond_wait(&c, &m);
}
mutex_unlock(&m);
```

PRODUCER/CONSUMER PROBLEM

EXAMPLE: UNIX PIPES

```
int pipefd[2]; // read from pipe[0], write to pipe[1]
if (pipe(pipefd) == -1) {
        perror("failed to create pipe");
        exit(EXIT_FAILURE);
}

// writer
char msgbuf[BUFSIZ];
snprintf(msgbuf, BUFSIZ, "Hello world\n");
write(pipefd[1], msgbuf, strlen(msgbuf) + 1);

// reader
int nbytes;
nbytes = read(pipefd[0], msgbuf, BUFSIZ - 1);
```

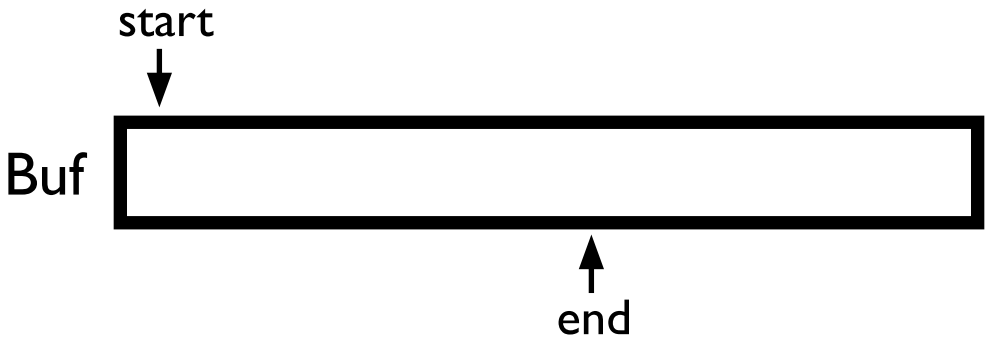
EXAMPLE: UNIX PIPES

A pipe may have many writers and readers

Internally, there is a finite-sized buffer

Writers add data to the buffer
- Writers have to wait if buffer is full

Readers remove data from the buffer
- Readers have to wait if buffer is empty



PRODUCER/CONSUMER PROBLEM

Producers generate data (like pipe writers)

Consumers grab data and process it (like pipe readers)

Producer/consumer problems are frequent in systems (e.g. web servers)

General strategy use condition variables to:

- make producers wait when buffers are full
- make consumers wait when there is nothing to consume

Producer/Consumer Example

We'll start with easy case:

- 1 producer thread
- 1 consumer thread
- 1 shared buffer to fill/consume (max = 1)

Numfull = number of slots currently filled

ATTEMPT#1

Global vars:

```
mutex_t  m;
cond_t   cv;
boolean done = 0;
```

Thread 1:

```
void *producer(void *arg) {
    while (true) {
        mutex_lock(&m);
        if (numfull == max)
            cond_wait(&cv, &m);
        produce_once();
        cond_signal(&cv);
        mutex_unlock(&m);
    }
}
```

Start with numfull = 0

Thread 2:

```
void *consumer(void *arg) {
    while (true) {
        mutex_lock(&m);
        if (numfull == 0)
            cond_wait(&cv, &m);
        int tmp = consume_once();
        cond_signal(&cv);
        mutex_unlock(&m);
        printf("%d\n", tmp);
    }
}
```

WHAT ABOUT 2 CONSUMERS?

Can you find a problematic timeline with 2 consumers (still 1 producer)?

```
void *producer(void *arg) {
    while (1) {
        mutex_lock(&m);          // p1
        if (numfull == max)      //p2
            cond_wait(&cv, &m);//p3
        produce_once();          // p4
        cond_signal(&cv);        //p5
        mutex_unlock(&m);        //p6
    }
}
```

```
void *consumer(void *arg) {
    while (1) {
        mutex_lock(&m);          // c1
        if (numfull == 0)        // c2
            cond_wait(&cv, &m); // c3
        int tmp = consume_once(); // c4
        cond_signal(&cv);        // c5
        mutex_unlock(&m);        // c6
    }
}
```

HOW TO WAKE UP THE CORRECT THREAD?

Wake all the threads!

```
pthread_cond_broadcast(cond_t *cv);
```

instead of

```
pthread_cond_signal(cond_t *cv);
```

Better solution (usually): use two condition variables

POP QUIZ #3

https://forms.gle/zgyPoR8wQa8rH5hcA

ATTEMPT#2: 2 CVs ARE BETTER THAN 1

```
void *producer(void *arg) {
    while (1) {
        mutex_lock(&m);          // p1
        if (numfull == max)      //p2
            cond_wait(&empty, &m);//p3
        produce_once();          // p4
        cond_signal(&fill);      //p5
        mutex_unlock(&m);        //p6
    }
}
```

```
void *consumer(void *arg) {
    while (1) {
        mutex_lock(&m);          // c1
        if (numfull == 0)        // c2
            cond_wait(&fill, &m); // c3
        int tmp = consume_once(); // c4
        cond_signal(&empty);     // c5
        mutex_unlock(&m);        // c6
    }
}
```

Solves the previous problem…
But can you still find a bad schedule?

Global vars:

```
mutex_t  m;
cond_t   full, empty;
boolean done = 0;
```

ATTEMPT#3: Two CVs and WHILE

```
void *producer(void *arg) {
    while (1) {
        mutex_lock(&m);             // p1
        while (numfull == max)      //p2
            cond_wait(&empty, &m);//p3
        produce_once();             // p4
        cond_signal(&fill);         //p5
        mutex_unlock(&m);           //p6
    }
}
```

```
void *consumer(void *arg) {
    while (1) {
        mutex_lock(&m);             // c1
        while (numfull == 0)        // c2
            cond_wait(&fill, &m); // c3
        int tmp = consume_once(); // c4
        cond_signal(&empty);        // c5
        mutex_unlock(&m);           // c6
    }
}
```

No concurrent access to shared state

Every time lock is acquired, assumptions are reevaluated

A consumer will get to run after every produce_once()

A producer will get to run after every consume_once()

GOOD RULE OF THUMB 3

Whenever a lock is acquired, recheck assumptions about state!

Another thread could grab lock in between signal and wakeup from wait

Note that some libraries also have “spurious wakeups” (may wake multiple waiting threads at signal or at any time)

Good stress test: change your signal to broadcast and see if your code still works

HOARE VS MESA SEMANTICS

Mesa: used widely (cheaper & easier to implement)

- Signal moves a waiting thread to READY/RUNNABLE state
- Signaler holds on to lock & the CPU/core
- Not necessary that the signaled waiter runs next

Conditions may change by the time waiter runs

Hoare: almost never used (expensive & complex to implement)

- Signal gives lock & the CPU/core to waiting thread
- Waiter runs when woken up by the signaler
- When waiter finishes, it hands lock + CPU/core back to signaler

Conditions remain unchanged because waiter runs immediately after

SUMMARY: 3 RULES OF THUMB FOR CVS

1. Need to keep some state in addition to CV's

2. Always wait or signal with lock held

3. Always recheck state when thread wakes up from a wait

   Use a while instead of an if

Synchronization

Generic synchronization: for all models from previous slides

Goal: Learn how synchronization primitives are built (lib + OS)
With a little help from our friend…the hardware

Monitors
Locks
Semaphores
Condition Variables

Atomic Load
XCGH, Test&Set, etc.
Atomic Store
Disable Interrupts

INTRODUCING Semaphores

Condition variables have no state (other than waiting queue)

- Programmer must track additional state

Semaphores have state: track integer value

- State cannot be directly accessed by user program, but state determines behavior of semaphore operations

Equivalence

Semaphores are equally powerful to Locks+CVs

- what does this mean?

One might be more convenient, but that's not relevant

Equivalence means each can be built from the other